The empirical recurrence for OEIS A232147, proved by transfer matrix

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Abstract

OEIS A232147 counts arrays of width 4 over $\{0, 1, 2\}$ in which every cell carrying one stated value must have a neighbour carrying another. The entry carries an empirical recurrence of order 32 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: a cell's neighbours lie in three consecutive rows, so the count is a walk count on $S = 6643$ states.

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1 The conjecture

OEIS A232147 is "Number of n X 4 0..2 arrays with every 0 next to a 1 and every 1 next to a 2 horizontally, diagonally or antidiagonally.". It has offset 1 and begins

$$15, 1299, 67725, 3409361, 177194147, 9181257593, 475203378037, 24603676419865, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 35*a(n-1) +649*a(n-2) +10479*a(n-3) +50649*a(n-4) -148850*a(n-5) -4594482*a(n-6) -29446017*a(n-7) -109126913*a(n-8) -189857827*a(n-9) +786606763*a(n-10) +4637371935*a(n-11) +5864860064*a(n-12) -12212688755*a(n-13) -42763674878*a(n-14) -28294485410*a(n-15) +57477014048*a(n-16) +126547355711*a(n-17) +64552653143*a(n-18) -60155915760*a(n-19) -96981273076*a(n-20) -64212382080*a(n-21) -123120194790*a(n-22) -55873185285*a(n-23) -22972924737*a(n-24) +79085877984*a(n-25) +44270889165*a(n-26) +95126815696*a(n-27) +10764163372*a(n-28) +18110185584*a(n-29) -9929666880*a(n-30) -6320643840*a(n-31) +1858775040*a(n-32).

As of the "Last modified" line on the live entry (May 11 2023 15:06:35, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 4 cells across, with entries in $\{0, 1, 2\}$. The NEIGHBOURS of a cell are the cells reached from it by the offsets

$$(-1, -1), (-1, 1), (0, -1), (0, 1), (1, -1), (1, 1),$$

those lying inside the array. The entry asks that

- every cell carrying 0 has at least one neighbour carrying 1;
- every cell carrying 1 has at least one neighbour carrying 2;

A cell whose value is not named on the left of any clause is unconstrained.

3 Three rows decide one

Lemma 1. *Every offset above changes the row index by -1, 0 or $+1$, so all the neighbours of a cell of row i lie in rows $i-1$, i and $i+1$, and the condition for the whole of row i is decided by those three rows.*

Proof. Immediate from the offsets listed, together with the rule that a neighbour outside the array is not a neighbour. □

So the state is the pair (row above, current row), the row above allowed to be the symbol $\perp$, and a step appends a row and settles the condition for the middle row of the window it completes. The condition for the LAST row is settled by the end vector, with nothing below it: the end vector is therefore not the all-ones vector, and a row whose cells find what they need only because a further row is written below them does not finish an array. With M the adjacency matrix,

$$a(n) \;=\; \iota^{\top} M^{\,j} \tau, \qquad S = 6643.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^{32} - \sum_i c_i t^{32-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+32} - \sum_i c_i M^{j+32-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 3.

$$a(n) \;=\; 35\,a(n-1)+649\,a(n-2)+10479\,a(n-3)+50649\,a(n-4)-148850\,a(n-5)-4594482\,a(n-6)-29446017\,a($$

for every $n > 31$.

Proof. The vector $w = q(M)\tau$ was formed by 32 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 31 + 1$ onwards and the run continued until the working vector was identically zero. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size in the orientation the entry's own name writes and test each cell against its neighbours directly. No window, no state and no transposition are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A232147.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.